DO NOW — RECALL

Lesson 131 · Asymmetric Encryption and Public-Key Use

Name: _____ Class: _____ Date: _____

1. What is the principal challenge of symmetric encryption between distant users?

2. What is a function parameter?

3. What is the binary result of 1 + 1 in a full adder with carry-in 0?

4. How many bits are in three bytes?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Securely sharing and managing the secret key.

2. Model answer

An input used by a subroutine.

3. Model answer

Sum 0, carry-out 1.

4. Model answer

24.

By the end, you can…

• Public key can be shared; private key remains secret.

• A message encrypted with recipient public key is decrypted by recipient private key.

• Signature created with private key is verified using corresponding public key.

Vocabulary — know these terms

public key — A key that can be distributed openly.

private key — A matching secret key kept by its owner.

asymmetric encryption — Uses a related public/private key pair.

digital signature — Cryptographic evidence of origin and integrity with private-key signing.

NEW LEARNING

Read, annotate and explain

Generate key pair

Public key can be shared; private key remains secret.

Encrypt to recipient

A message encrypted with recipient public key is decrypted by recipient private key.

Sign and verify

Signature created with private key is verified using corresponding public key.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

Bob shares public key

Predict the first step from the prompt.

Alice encrypts for Bob

Explain the intermediate result or decision.

Bob uses private key

Alice encrypts a suitably sized message or session key using Bob's public key. Bob decrypts using his matching private key. Alice never requires Bob's private key.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the difference between public and private keys and how they support confidentiality. [6]

Question 2

Outline how a digitally signed download is verified. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare asymmetric and symmetric approaches for sending a large encrypted video file. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A public key must be kept secret for public-key encryption to work.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.